

Project 1: Market Quantitative Analysis

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Target

The following task requires the candidate to demonstrate their ability to apply quantitative finance techniques within the context of commodity markets, specifically coffee. The individual is asked to address three key areas:

- **Pricing a Futures Contract** – The candidate should apply the cost of carry model to determine the fair value of a coffee futures contract, ensuring that the price reflects the spot value, storage costs, the risk-free interest rate, and time to maturity.
- **Structuring Securities** – The candidate should outline how to design securities linked to coffee prices, such as commodity-linked bonds or structured notes. This involves identifying the statistical relationship between coffee prices and economic factors, while incorporating stochastic processes (e.g., geometric Brownian motion) to simulate price paths and determine payout structures.
- **Managing Risk** – The candidate should propose quantitative methods to mitigate risks in coffee-related investments. This includes techniques such as Value at Risk (VaR) to estimate potential losses, Monte Carlo simulations to explore multiple price scenarios, and GARCH models to forecast volatility and assess its impact on portfolio performance.

By addressing these points, the candidate demonstrates not only technical proficiency in pricing, structuring, and risk management, but also the ability to integrate theory with practical applications in financial markets.

Script

The following Python script builds a structured product linked to coffee futures prices. It combines a zero-coupon bond (capital protection) with a call option (upside participation). Each block is explained below:

Defined parameters. This section establishes the key inputs: the current futures price, strike price, risk-free rate, volatility, maturity, and total invested capital. These parameters drive all subsequent calculations.

```
# Parametros
F0 = 200          # Precio del futuro del cafe
K = 200           # Strike de la opcion
r = 0.05          # Tasa libre de riesgo
sigma = 0.20      # Volatilidad
T = 1             # Tiempo en anos
capital_total = 1000 # Monto total invertido (ej. $1000)
```

Present value of the zero-coupon bond. Here, the capital protection component is secured by investing in a zero-coupon bond, discounted at the risk-free rate. This ensures the investor recovers at least the principal at maturity.

```
PV_bono = capital_total * np.exp(-r * T)
```

Black School model. The Black model is applied to price the call option on coffee futures. The values of $d1$ and $d2$ are calculated, followed by the option premium, which reflects the cost of participation in the upside of coffee prices.

```
d1 = (np.log(F0 / K) + 0.5 * sigma ** 2 * T) / (sigma * np.sqrt(T)
→ ))
d2 = d1 - sigma * np.sqrt(T)
N_d1 = norm.cdf(d1)
N_d2 = norm.cdf(d2)
call_price = np.exp(-r * T) * (F0 * N_d1 - K * N_d2)
```

Budget for options. The remaining capital, after purchasing the bond, is allocated to buy call options. The script computes how many options can be acquired with this budget.

```
presupuesto_opciones = capital_total - PV_bono
cantidad_opciones = presupuesto_opciones / call_price
```

Simulate the final value of the structured product based on future coffee scenarios. Using a range of possible coffee futures prices, the script simulates the product's payoff. The investor participates in 50% of the upside while the capital protection remains intact.

```
precios_futuros = np.linspace(150, 250, 100)
ganancia_opcion = np.maximum(precios_futuros - K, 0)
retorno_cliente = PV_bono + cantidad_opciones * 0.5*
→ ganancia_opcion # Participacion del 50%
```

Print the result. Key financial outputs are displayed: the bond's present value, the option price, the number of options purchased, and the option budget. These results help quantify the structure's balance between risk and return.

```
print("Valor presente del bono cupon cero (proteccion capital): $
→ {:.2f}".format(PV_bono))
print("Precio de la call (modelo Black): $ {:.2f}".format(
→ call_price))
print("Cantidad de opciones compradas: {:.2f}".format(
→ cantidad_opciones))
print("Presupuesto para opciones: $ {:.2f}".format(
→ presupuesto_opciones))
```

Graph of the value of the structured product. Finally, the payoff profile of the structured product is plotted. The chart shows how the investment behaves under different coffee price scenarios, highlighting the capital protection line and the upside participation.

```
plt.figure(figsize=(10,6))
plt.plot(precios_futuros, retorno_cliente, label='Valor
→ estructurado al vencimiento', color='green')
plt.axhline(capital_total, color='gray', linestyle='--', label='
→ Capital protegido')
plt.xlabel('Precio del Futuro del Cafe al Vencimiento')
plt.ylabel('Valor del Producto Estructurado ($)')
```

```
plt.title('Simulación de Producto Estructurado con Participación  

→ del 50% en la Subida del Café')
plt.legend()
plt.grid(True)
plt.show()
```

Results

Based on the quantitative models, the following outcomes were obtained:

1. Present value of the zero-coupon bond (capital protection): \$951.23.
2. Call price (Black model): \$15.15.
3. Number of options purchased: \$3.22.
4. Option budget: \$48.77



Figure 1: Simulation of a Structured Product — resultado.

Conclusion

From the analysis of the structured product, the following conclusions can be drawn:

1. The combination of a zero-coupon bond and call options provides an effective balance between capital protection and upside participation.
2. The use of the Black model allows for a fair valuation of options, offering transparency in the cost of derivative exposure.
3. The residual budget determines the investor's capacity to enhance returns through derivative instruments, highlighting the importance of efficient capital allocation.

4. Simulations across different coffee futures prices show that the product consistently safeguards principal while granting partial exposure to favorable market movements.
5. This approach illustrates how structured products can be tailored to investor risk preferences, blending security with growth potential.